

Network Design and Engineering (CST 4404)

Review Questions:

R1. Network design is a critical process that it can meet the needs of an organization. Explain the **main attributes of network design** that define how a network should be designed and implemented.

(1) Performance

- **Bandwidth:** A good network design should have enough bandwidth to meet the needs of users and applications without congestion.
- **Latency:** The network should be designed to minimize data transmission delay, ensuring that applications that require real-time data (like video conferencing or online gaming).
- **Throughput:** This is the actual speed at which data is transmitted in the network.

Throughput should be maximized to handle large amounts of data efficiently.

(2) Scalability

- **Future updates:** The network should be designed so that it can grow as the organization's needs expand. This includes adding new devices, users, or applications without significant redesign or performance degradation.
 - Eg: change interfaces from copper to fiber based physical connections

(3) Reliability

- **Fault Tolerance:** The network should be able to continue functioning even if one or more components fail. Redundant links, backup power supplies, and failover mechanisms should be in place.

- **High Availability:** This refers to the network's uptime or how often it is operational. A well-designed network ensures high availability, **minimizing downtime**.
- **Recovery:** The design should include disaster recovery plan and mechanisms to restore services in case of failure.

(4) Security

- **Physical security, Services security:** A secure network design protects the organization's data, devices, and resources from unauthorized access, attacks, and breaches.
- Security measures might include firewalls, encryption, intrusion detection systems (IDS), access control policies, and secure authentication methods.

(5) Manageability

- **Network management tools and systems:** should be part of the design to monitor and control the network efficiently.
- The design should facilitate easy network monitoring, troubleshooting, configuration changes, and updates. This includes the use of centralized management solutions, automation, and easy-to-use interfaces.

(6) Quality of Service (QoS)

- **Treatment of traffic, Real time versus non-real time traffic:** QoS is a critical attribute for ensuring that network traffic is prioritized according to its importance.
 - Example, voice or video traffic may need to be prioritized over standard data traffic to maintain the quality of communication.
- The network design should ensure that mission-critical applications or

services always get the necessary resources and bandwidth to function properly.

(7) Topology

- Physical and logical layout of the network: is crucial for both performance and manageability. The design should select an appropriate network topology (e.g., star, mesh, bus, etc.) based on the size, requirements, and goals of the network.
- The topology influences factors like fault tolerance, scalability, and performance.

R2. Hierarchical networks use a tiered design of each layer performing a well-defined role in the campus network. Explain the **role of each layer in the three-tier architecture**. / How does a **three-tier system** handle redundancy, security and performance? / How does **a three-tier architecture handle failover and redundancy**?

Each layer in a three-tier architecture perform as follows

- ✓ The access layer represents the network edge, where traffic enters or exits the campus network. An access layer switch is to provide network access to the user. To meet network application and end-user demand, the next-generation switching platforms provide more converged, integrated, and intelligent to operate on the network more efficiently and securely.
- ✓ The distribution layer interfaces between the access layer and the core layer to provide many important functions:
 - Aggregating Layer 2 broadcast domains and Layer 3 routing
 - Providing intelligent switching, routing, and network access policy

- Providing high availability through redundant distribution layer switches
 - Providing differentiated services to various classes of service applications at the edge of the network.
- ✓ The core layer is the network backbone. It serves as the aggregator for all of the distribution layer devices and ties the campus together with the rest of the network. The primary purpose of the core layer is to provide fault isolation and high-speed backbone connectivity.

R3. What are the **characteristic of scalable** network architecture that can grow without losing availability and reliability? / What are the considerations for designing a scalable network?

When designing a scalable network, the following criteria should consider:

- ✓ Redundant Links
- ✓ Multiple Links with Link aggregation (EtherChannel)
- ✓ Scalable Routing Protocol
- ✓ Reduce Failure domain size
- ✓ Switch block deployment

R4. Explain the basic concept and the **primary responsibilities of data plane and control plane** in a network device.

- ✓ The data plane functions of the network layer determine how a datagram arriving on one of a router's input links is forwarded to one of that router's output links.
- ✓ The control plane functions of the network layer that controls how a

datagram is routed among routers along an end-to-end path from the source host to the destination host.

R5. Explain the **two main functions** of the network layer devices.

- ✓ **Forwarding** is about moving a packet from a router's input link to the appropriate output link.
- ✓ **Routing** is about determining the end-to-end routes between sources and destinations.

R6. Where **can queueing occur** in a router? Briefly explain the conditions that lead to such queueing.

Queueing can occur at both the input ports and the output ports of a router.

- ✓ Queue occurs **on an input port** when the arriving rate of packets exceeds the switch capacity;
- ✓ Queueing **occurs at the output port** when datagrams arrive from fabric faster than the transmission rate of output line speed.

R7. Why **HOL blocking occur**?

- ✓ HOL blocking a **queued packet in an input queue must wait for transfer through the fabric** because it is blocked by another packet at the head of the line.
- ✓ It occurs at the input port.

R8. What is the advantage of switching via an interconnection network (crossbar switching type) over switching via bus?

- ✓ A crossbar switch is an interconnection network that **connects N input ports to N output ports using 2N buses.**
- ✓ Therefore, this technique overcome the bandwidth limitation of a single shared bus.

R9. Discuss the basic **difference between Intra-AS and Inter-AS routing** and give an example of protocol used in both.

Intra-AS (routing within a single Autonomous System):

- ✓ All routers in AS must run same intra-domain protocol
- ✓ Routers in different AS can run different intra- domain routing protocols, eg:

Open Shortest Path First Protocol (OSPF)

Inter-AS (routing between multiple Autonomous Systems)

- ✓ Gateways perform inter-domain routing (as well as intra-domain routing)
- ✓ eg: **Border Gateway Protocol (BGP)**

R10. How does **OSPF handle routing between multiple areas in an AS** and how the routing information is exchanged between areas?

- ✓ In OSPF, routing between multiple areas is handled primarily by **Area Border Routers (ABRs).**
- ✓ ABRs **summarize and forward routing information** between areas, ensuring that routers in different areas can communicate.
- ✓ Routing within an area rely on link-state information, while routing between areas involves summary LSAs and the backbone (Area 0) serving as the central point for inter-area traffic.

R11. Explain the **purpose of BGP** in managing routing between different Autonomous Systems. How it **handles path selection**?

BGP is an Exterior Gateway Protocol (EGP) used to exchange routing information between different Autonomous Systems.

Its main purpose is to:

- ✓ **Inter-AS Routing:** Border Gateway Protocol (BGP) enables routers to exchange routing information across different ASes.
- ✓ **Path Control and Policy-based Routing:** BGP allows network administrators to **implement routing policies** based on business needs.
- ✓ **Scalability:** BGP is designed to scale efficiently across a large number of ASes, handling a significant volume of routing information (the Internet's routing table).

It uses a path vector mechanism, where each BGP router knows about the entire path to a destination, rather than just the next-hop router.

- ✓ **Loop Prevention:** One of the most critical features of BGP is its ability to prevent routing loops between ASes. It achieves this using the AS Path attribute, which records the ASes that a route has traversed.

BGP uses several key attributes to determine the best path to a destination. Two key attributes are:

(1) AS Path

- ✓ AS Path attribute is a list of ASes that the route has traversed. The AS Path is **primarily used for loop prevention. If a router sees its own AS number in the AS Path, it knows that the route has already passed through its AS, indicating a potential loop.**

- ✓ It Affects Path Selection: BGP prefers **shorter AS Paths**, i.e., fewer ASes in the path. If **multiple paths exist to the same destination**, BGP will **choose the one with the shortest AS Path**.

- Example:

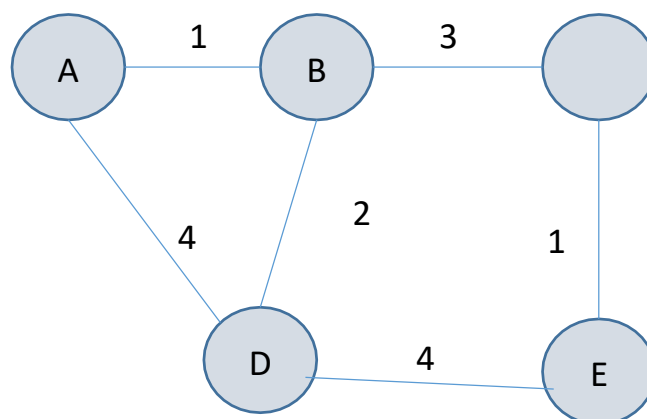
- Route 1: AS 100 → AS 200 → AS 300
- Route 2: AS 100 → AS 150 → AS 200
- BGP will prefer Route 2, as it has a shorter AS Path.

(2) Next-Hop

- ✓ Next-Hop attribute indicates the **IP address of the next router** that the destination can be reached through. It is critical for determining where to send traffic. It tells the router which next-hop IP address to use to reach a particular destination network.

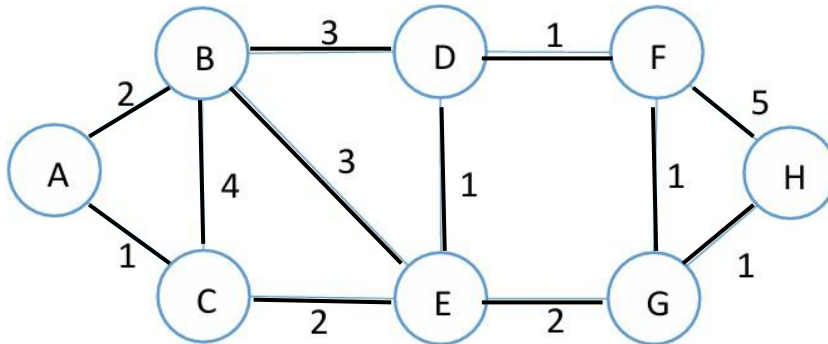
R12. Apply **Dijkstra's Link state routing algorithm** to **find the shortest paths from node A to all other nodes in a graph**, with the given link costs.

- (a) Draw a table showing the distance values of all the nodes at all intermediate steps. (b) Show the final shortest path tree from node A to all other nodes.



R13. Apply Dijkstra's Link state routing algorithm to find the shortest paths from node A to all other nodes in a graph, with the given link costs.

- (a) Draw a table showing the distance values of all the nodes at all intermediate steps. (b) Show the final shortest path tree from node A to all other nodes.



R14. What is an Access Control List (ACL) in networking?

- ✓ It is a set of rules that control network traffic.
- ✓ It defines whether to permit or deny packets based on source and destination IP addresses, protocol types, and other packet attributes.
- ✓ ACLs are used to enhance security by filtering traffic in a network.

R15. What are the common types of ACL? There are two primary types of ACLs.

- ✓ Standard ACLs: Filter traffic based on the source IP address only.
- ✓ Extended ACLs: Filter traffic based on source/destination IP addresses, protocols and port numbers.

R16. Where should you apply a standard ACL in a network?

- ✓ Standard ACLs should be applied as **close to the destination** as possible because they **filter traffic based only on the source IP address**.
- ✓ Applying them too close to the source might unintentionally block legitimate traffic.

R17. Where should you apply an extended ACL in a network?

- ✓ Extended ACLs should be applied as **close to the source** as possible.
- ✓ This **minimizes unnecessary traffic** in the network by filtering out unwanted traffic before it traverses the network.

R18. What is the difference between a numbered ACL and a named ACL?

- ✓ Numbered ACL: An ACL identified by a number (1-99 for standard, 100-199 for extended).
- ✓ Named ACL: An ACL identified by a descriptive name. Named ACLs provide more flexibility as they allow adding or removing specific rules without removing the entire ACL.

R19. What is the implicit "deny all" rule in ACLs?

- ✓ There is an implicit "deny all" rule at the end of every ACL.
- ✓ This means that if a packet does not match any of the permit rules in the ACL, it will be denied by default.

R20. How does an ACL process traffic?

- ✓ It is in a sequential manner, from top to bottom.
- ✓ When a packet matches a rule, the ACL stops processing further rules.
- ✓ If the packet does not match any rule, the implicit "deny all" rule applies.

R21. What is the difference between "inbound" and "outbound" ACLs?

- ✓ Inbound ACL: Filters traffic entering an interface before it is routed.
- ✓ Outbound ACL: Filters traffic after it has been routed, as it exits the interface.

R22. How do ACLs enhance network security?

- ✓ ACLs enhance network security **by restricting unauthorized access, controlling the flow of traffic**, and preventing specific types of traffic from entering or leaving a network.
- ✓ They can be used to **prevent attacks, limit access to critical resources**, and **enforce network policies**.

R23. What is the **impact** of incorrectly configured ACLs?

- ✓ Incorrectly configured ACLs **can block legitimate traffic, leading to network outages or degraded performance**.
- ✓ They can also **fail to block malicious traffic**, leaving the network vulnerable to attacks.

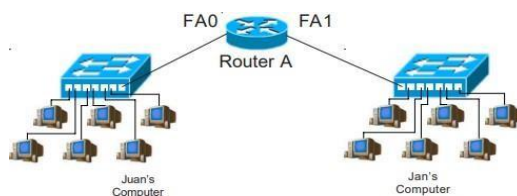
R24. Write the Standard ACL for Basic Traffic Filtering. Block traffic from a specific host. Create a standard ACL (numbered 1-99) to deny traffic from host 192.168.1.10. Apply the ACL to inbound traffic on interface GigabitEthernet0/0.

- ✓ access-list 10 deny host 192.168.1.10
- ✓ access-list 10 permit any
- ✓ interface GigabitEthernet0/0
- ✓ ip access-group 10 in

R25. Write the extended ACL for Specific Services. Allow HTTP/HTTPS traffic to a web server. Create an extended ACL (numbered 100-199) to permit HTTP/HTTPS to 10.0.0.5. Apply the ACL to outbound traffic on interface GigabitEthernet0/.

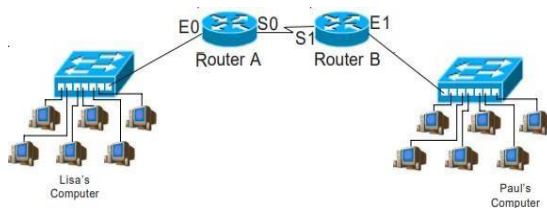
- ✓ access-list 110 permit tcp any host 10.0.0.5 eq 80
- ✓ access-list 110 permit tcp any host 10.0.0.5 eq 443
- ✓ access-list 110 deny ip any any

R26. In order to permit packets from Juan's computer to arrive at Jan's computer you would place the standard access list at router interface



Answer: FA1

R27. Lisa has been sending unnecessary information to Paul.



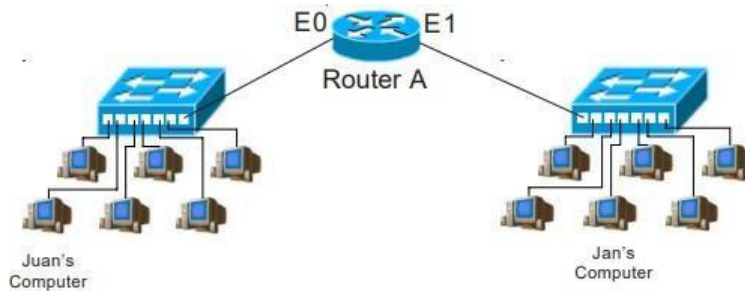
- ✓ Where would you place the standard ACL to deny all traffic from Lisa to Paul?

Answer: Router B, E1

- ✓ Where would you place the standard ACL to deny traffic from Paul to Lisa?

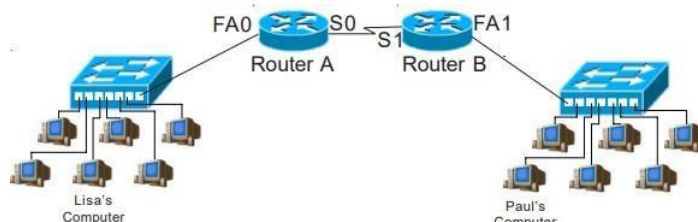
Answer: Router A, E0

R28. In order to permit packets from Juan's computer to arrive at Jan's computer you would place the extended access list at router interface



a. Answer: E0

R29. Lisa has been sending unnecessary information to Paul. Where would you place the extended ACL to deny all traffic from Lisa to Paul?

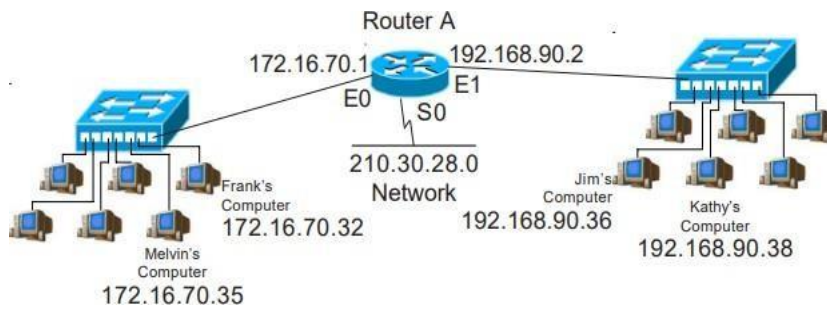


Answer: Router A, FA0

✓ Where would you place the extended ACL to deny traffic from Paul to Lisa?

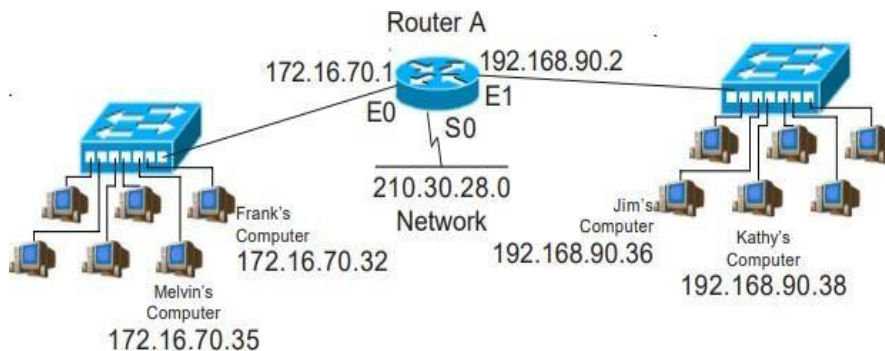
Answer: Router B, FA1

R30. Write a named standard access list called "GEO" on Router A, interface E1 to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic. Place the access list at



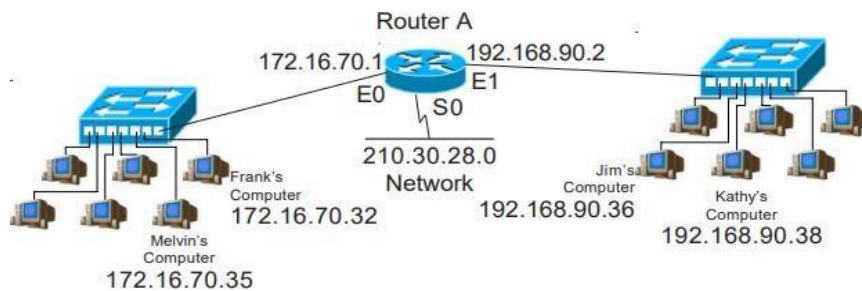
- ✓ ip access-list standard GEO
- ✓ deny host 172.16.70.35
- ✓ access-list permit any
- ✓ interface E1
- ✓ ip access-group GEO out

R31. Write a standard access list to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Place the access list at:



- ✓ access-list 10 deny host 172.16.70.35
- ✓ access-list 10 permit any
- ✓ interface E1
- ✓ ip access-group 10 out

R32. Write a standard access list to block Jim's computer from sending information to Frank's computer; but will allow all other traffic from the 192.168.90.0 network. Permit all traffic from the 210.30.28.0 network to reach the 172.16.70.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Place the access list at:



- ✓ access-list 25 deny 192.168.90.36
- ✓ access-list 25 permit 192.168.90.0 0.0.0.255
- ✓ access-list 25 permit 210.30.28.0 0.0.0.255
- ✓ interface E0
- ✓ ip access-group 25 out

R33. Create an ACL that allows HTTP (TCP port 80) traffic from the source network 192.168.11.0/24 to the destination network 192.168.20.0/24. Deny all other traffic from 192.168.11.0/24 to 192.168.20.0/24. Apply the ACL to the correct interface.

- ✓ access-list 100 permit tcp 192.168.11.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 80
- ✓ access-list 100 deny ip 192.168.11.0 0.0.0.255 192.168.20.0 0.0.0.255
- ✓ access-list 100 permit ip any any

R34. Create an extended ACL to allow only ICMP and HTTP traffic from 192.168.20.0/24 to 192.168.40.0/24. Deny all other traffic.

- ✓ access-list 101 permit icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
- ✓ access-list 101 permit tcp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 eq 80
- ✓ access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
- ✓ access-list 101 permit ip any any

R35. Create an extended ACL that denies all traffic from the source IP address 192.168.50.10 to any destination, but allows all other traffic.

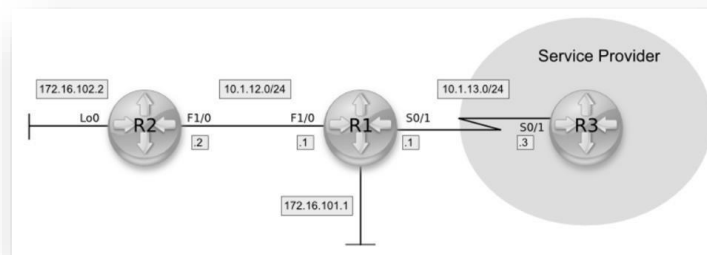
- ✓ access-list 102 deny ip host 192.168.50.10 any
- ✓ access-list 102 permit ip any any

R36. Create an ACL that allows DNS (port 53), HTTP (port 80), and HTTPS (port 443) traffic from the network 192.168.100.0/24 to any destination but denies all other traffic.

- ✓ access-list 105 permit udp 192.168.100.0 0.0.0.255 any eq 53
- ✓ access-list 105 permit tcp 192.168.100.0 0.0.0.255 any eq 80
- ✓ access-list 105 permit tcp 192.168.100.0 0.0.0.255 any eq 443
- ✓ access-list 105 deny ip 192.168.100.0 0.0.0.255 any
- ✓ access-list 105 permit ip any any

R37. You are in charge of R1 and R2 routers. R3 belongs to your Service Provider network and simulates Internet.

(i) Configure an Extended access-list disabling anyone TELNET to R1 and all



devices behind it (R2) if the traffic is originated from Internet (here: SP). All other traffic should be permitted.

- access-list 100 deny tcp any any eq telnet
- access-list 100 permit ip any any
- interface s0/1
- ip access-group 100 in

(ii) On R1 remove the previous ACL and configure a new one allowing only HTTP access to 172.16.102.0/24 if the traffic is originated from Internet (here: SP). All other traffic should be discarded.

- ✓ no access-list 100
- ✓ interface s0/1
- ✓ no ip access-group 100 in
- ✓ access-list 101 permit tcp any host 172.16.102.2 eq www
- ✓ interface s0/1
- ✓ ip access-group 101 in

R38. Write the Wildcard Mask for specific service. Allow only the 192.168.4.0/24 subnet. Use a wildcard mask to match the subnet:

✓ access-list 20 permit 192.168.4.0 0.0.0.255

R39. Compare Static NAT, Dynamic NAT, and PAT (Port Address Translation).

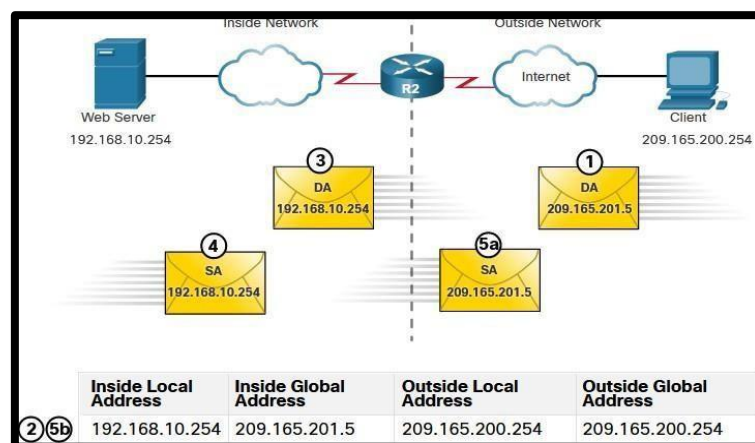
Static NAT	Dynamic NAT	PAT
one-to-one mapping of local and global addresses	one-to-one mapping of local and global addresses	One Inside Global address (public) can be mapped to many Inside Local addresses (private).
A unique Inside Global address is required for each inside host accessing the outside network.	A unique Inside Global address is required for each inside host accessing the outside network. But it can assign an available public IPv4 address from the pool	A single unique Inside Global address can be shared by many inside hosts accessing the outside network.

R40. How does NAT work and why is it used in networking?

✓ NAT allow a device with a private IPv4 address to access devices and resources outside of the local network, the private address must first be translated to a public address. NAT provides the translation of private addresses to public addresses.

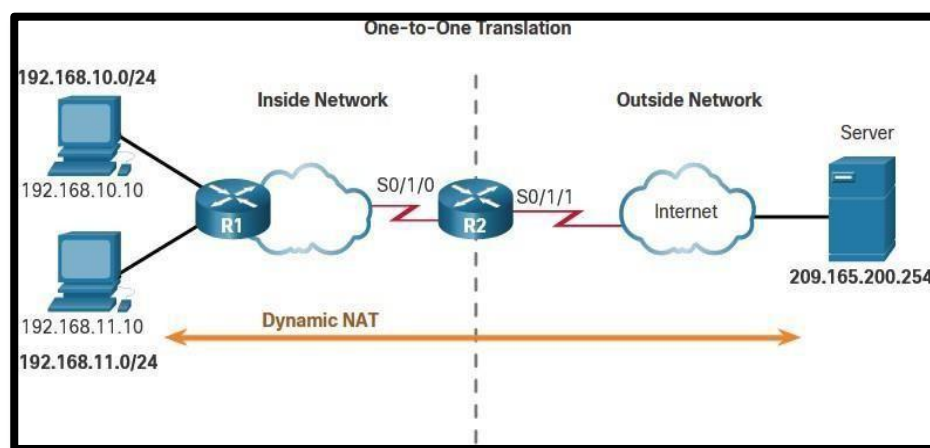
- ✓ NAT conserves the legally registered public addressing scheme by allowing the privatization of intranets. It conserves addresses through application port-level multiplexing. NAT provides consistency for internal network addressing schemes. It hides the IPv4 addresses of users and other devices in a local network. It helps with IP address conservation, security, and stable network design.

R41. Static NAT allows external devices to initiate connections to internal devices using the statically assigned public address. In the figure, an internal web server may be mapped to a specific inside global address so that it is accessible from outside networks. Analyze how the static NAT translation works between client host 209.165.200.254 and Web Server 192.168.10.254.



- (1) The client sends a packet to the web server.
- (2) R2 receives packets from the client on its NAT outside interface and checks its NAT table.
- (3) R2 translates the inside global address to the inside local address and forwards the packet towards the web server.
- (4) The web server receives the packet and responds to the client using its inside local address.
- (5) R2 receives the packet from the web server on its NAT inside interface with source address of the inside local address of the web server and translates the source address to the inside global address.

R42. In the following scenario, dynamic NAT uses a pool of inside global address 209.165.200.226 and 209.165.200.227. Analyze how Dynamic NAT works between client hosts (192.168.10.10, 192.168.11.10) and Server (209.165.200.254).

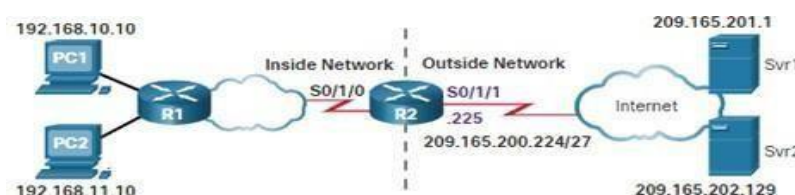


- (1) PC1 and PC2 send packets requesting a connection to the server.
- (2) R2 receives the first packet from PC1, checks the ALC to determine if the packet should be translated, selects an available global address, and creates a translation entry in the NAT table.

- (3) R2 replaces the inside local source address of PC1, 192.168.10.10, with the translated inside global address of 209.165.200.226 and forwards the packet. (The same process occurs for the packet from PC2 using the translated address of 209.165.200.227.)
- (4) The server receives the packet from PC1 and responds using the destination address of 209.165.200.226. The server receives the packet from PC2, it responds to using the destination address of 209.165.200.227.
- (5) When R2 receives the packet with the destination address of 209.165.200.226; it performs a NAT table lookup and translates the address back to the inside local address 192.168.10.10 and forwards the packet toward PC1.

When R2 receives the packet with the destination address of 209.165.200.227; it performs a NAT table lookup and translates the address back to the inside local address 192.168.11.10 and forwards the packet toward PC2. PC1 and PC2 receive the packet and continue the conversation.

R43. In the following scenario, Port address translation (PAT) uses different port numbers 1444 and 1445 to share a single public IP address 209.165.200.225 and distinguish between connections. All hosts from network 192.168.0.0/16 (matching ACL 1) that send traffic through router R2 to the internet. Analyze how PAT works between client hosts (192.168.10.10, 192.168.11.10) and Servers (209.165.201.1, 209.165.202.129).



- (1) PC1 and PC2 send packets to Svr1 and Svr2.
- (2) The packet from PC1 reaches R2 first. R2 modifies the source IPv4 address to 209.165.200.225 (inside global address). The packet is then forwarded towards Svr1.
- (3) The packet from PC2 arrives at R2. PAT changes the source IPv4 address of PC2 to the inside global address 209.165.200.225. PC2 has the same source port number as the translation for PC1. PAT increments the source port number until it is a unique value in its table. In this instance, 1445.
- (4) The servers use the source port from the received packet as the destination port, and the source address as the destination address for the return traffic.
- (5) R2 changes the destination IPv4 address of the packet from Srv1 from 209.165.200.225 to 192.168.10.10 and forwards the packet toward PC1. R2 changes the destination address of packet from Srv2 from 209.165.200.225 to 192.168.10.11. and modifies the destinations port back to its original value of 1444. The packet is then forwarded toward PC2.

R44. What is the single point of failure (SPOF) issue in a hub-and-spoke network design, and how would you mitigate it? / How does dual-homing address SPOF in a hub-and-spoke network, and what are the benefits?

- ✓ The hub in a hub-and-spoke topology is the **single point of failure**. If the hub goes down, the communication between all spokes is disrupted.
- ✓ To mitigate this risk, we could use **dual-homing** to connect **both the hub and spokes to multiple devices or links**. This ensures that if one link or device fails, traffic can still be rerouted through the secondary path.

- ✓ Dual-homing involves connecting both the hub and the spokes to two independent devices or links. This provides redundancy, as traffic can be rerouted if one link or device fails.

Benefits:

- Redundancy: Allow automatic failover in case of failure.
- Improved Availability: In case one link fails, the other one can take over without significant network downtime.

R45. Define Virtual Private Network (VPN). What are the **major benefits of VPN** and why is it important?

- ✓ A VPN is a technology that allows a secure connection over a public network (such as the internet) by using encryption and tunneling protocols.
- ✓ VPNs are important for privacy, secure remote access, data integrity, and to bypass geo-restrictions.
- ✓ Major benefits of VPNs are cost savings, security, scalability, and compatibility.

R46. What is IPsec and how does it work in a VPN?

- ✓ IPsec is an IETF standard that defines how a VPN can be secured across IP networks. IPsec protects and authenticates IP packets between source and destination.
 - It operates in two modes:
 - Transport Mode: Only the payload (data) is encrypted.
 - Tunnel Mode: The entire IP packet is encrypted and encapsulated within a new IP packet, which is used for site-to-site VPNs.

- ✓ IPsec uses various protocols, such as AH (Authentication Header) for integrity and ESP (Encapsulating Security Payload) for encryption.

R47. How does encryption work in an IPsec VPN, and what algorithms are typically used?

- Encryption in IPsec:
 - IPsec ensures data confidentiality by encrypting the data using algorithms like AES (Advanced Encryption Standard) or 3DES (Triple DES).
 - The data is encrypted and encapsulated in a secure tunnel between the two VPN endpoints.

Commonly Used Algorithms:

- AES (128, 192, or 256-bit): Strong encryption standard for modern IPsec VPNs.
- 3DES: Older and less secure than AES but still used in some legacy systems.
- DES (Data Encryption Standard): Deprecated and not recommended due to its vulnerability.

R48. What is an authentication header (AH) in IPsec, and how does it differ from encapsulating security payload (ESP)?

- AH (Authentication Header):
 - AH provides data integrity, authentication for IP packets.
 - AH does not provide encryption of the payload, meaning it only ensures that the data has not been altered during transmission.
- ESP (Encapsulating Security Payload):
 - ESP provides both data encryption and data integrity.

R49. What is FIFO (First-In, First-Out) Queuing, and how does it work?

- ✓ Packets wait in buffer (queue) until the node (router or switch) is ready to process them.
- ✓ If average arrival rate is higher than average processing rate, then queue will fill up and new packets will be discarded.
- ✓ FIFO has no concept of priority or classes of traffic and consequently, makes no decision about packet priority.
- ✓ There is only one queue, and all packets are treated equally.
- ✓ Packets are sent out an interface in the order in which they arrive. What are the drawbacks of FIFO in a congested network?

FIFO doesn't account for packet importance, meaning that high-priority traffic could be delayed due to a queue of lower-priority traffic. It can cause latency for real-time applications like VoIP or video conferencing.

In what scenarios would FIFO be preferred?

- ✓ FIFO works well in low-traffic networks or networks that do not require strict QoS controls, such as internal systems where no prioritization is necessary.

R50. What is Priority Queuing (PQ), and how does it work? What are the advantages of using Priority Queuing? What are the disadvantages of Priority Queuing? Give an example where Priority Queuing would be beneficial.

- ✓ In Priority Queuing, packets are first assigned to a priority class. Each priority class has its own queue.
- ✓ The packets in the higher-priority queue are processed first.
- ✓ Packet in the lowest-priority queue are processed last. A priority queue can provide better QoS than the FIFO queue. Higher- priority traffic (e.g., multimedia) can reach the destination with less delay. If there is a

continuous flow in a high-priority queue, the packets in the lower-priority queues will never have a chance to be processed. This is a condition called starvation.

The advantages of using Priority Queuing:

- ✓ Low-latency for high-priority traffic: Priority Queuing ensures that critical traffic (like VoIP or real-time video) is processed first.
- ✓ Useful in situations where latency-sensitive applications need to be prioritized.

The disadvantages of Priority Queuing

- ✓ Starvation: Low-priority packets may never get processed if high-priority queues are always full, leading to packet loss.

An example where Priority Queuing would be beneficial, VoIP or real-time video conferencing traffic can be given priority to reduce latency and ensure high-quality communication.

R51. What is Weighted Fair Queuing (WFQ), and how does it differ from FIFO and Priority Queuing?

- ✓ Packets are assigned to different classes and admitted to different queues.
- ✓ Weight are assigned to each queue based on their priority.
- ✓ Higher priority means a higher weight.
- ✓ WFQ can be seen as a combination of FIFO and Priority Queuing, but with fairer treatment of all traffic classes.

R52. How does the Token Bucket algorithm differ from the Leaky Bucket algorithm? / Explain the Leaky Bucket algorithm and how it works? / Explain the token bucket algorithm and how it works?

- ✓ The Token Bucket algorithm is more flexible than the Leaky Bucket. In

this model:

- Tokens are added to the bucket at a constant rate, and each packet requires a token to be transmitted. If there are tokens in the bucket, packets can be sent; if not, they have to wait.
- Unlike the Leaky Bucket, the Token Bucket allows bursts of traffic as long as there are enough tokens in the bucket without dropping packets.
- In Leaky bucket, data packets are added to the bucket at a constant rate (e.g., once every packet arrives). The bucket leaks data out at a constant rate, meaning the data is transmitted at a fixed rate. If the incoming traffic exceeds the rate at which the bucket leaks, the bucket overflows, and packets are dropped.

R53.What is the Integrated Services (IntServ) model, and how does it work? What are the advantages and limitations of using IntServ?

In the IntServ model, the application requests a specific kind of service from the network before sending data. IntServ uses the Resource Reservation Protocol (RSVP) to signal the QoS needs of an application's traffic along devices in the end-to-end path through the network.

Benefits	Drawbacks
▪ Explicit end-to-end resource admission control ▪ Per-request policy admission control ▪ Signaling of dynamic port	▪ Resource intensive due to the stateful architecture requirement for continuous signaling. ▪ Flow-based approach not scalable to large implementations such as the

numbers

internet.

R54.What is the differentiated services (DiffServ) QoS model, and how does it work? What are the advantages and limitations of using DiffServ?

- ✓ DiffServ divides network traffic into classes based on business requirements. Each of the classes can then be assigned a different level of service. As the packets traverse a network, each of the network devices identifies the packet class and services the packet according to that class.

Benefits	Drawbacks
✓ Highly scalable ✓ Provides many different levels of quality	✓ No absolute guarantee of service quality ✓ Requires a set of complex mechanisms to work in concert throughout the network

R55.What are the key components of SNMP? Discuss SNMP Manager, SNMP Agent, MIB (Management Information Base)/ Explain how an SNMP manager interacts with network devices and retrieves data.

SNMP system consists of three elements:

- SNMP manager
- SNMP agents (managed node)
- Management Information Base (MIB)

- ✓ SNMP manager is part of a network management system (NMS). The SNMP manager runs SNMP management software. SNMP manager can collect information from an SNMP agent by using the “get” action and can change configurations on an agent by using the “set” action. In addition, SNMP agents can forward information directly to a network manager by using “traps”.
- ✓ The SNMP agent and MIB reside on SNMP client devices. Network devices that must be managed, such as switches, routers, servers, firewalls, and workstations, are equipped with an SNMP agent software module. MIBs store data about the device and operational statistics and are meant to be available to authenticated remote users. The SNMP agent is responsible for providing access to the local MIB.

R56. Why is time synchronization important in a network? / Discuss the significance of time synchronization in a network.

- ✓ The software clock on a router or switch starts when the system boots. It is the primary source of time for the system. It is important to synchronize the time across all devices on the network.
- ✓ When the time is not synchronized between devices, it will be impossible to determine the order of the events and the cause of an event. Typically, the date and time settings on a router or switch can be set by using manually configure or configure the Network Time Protocol (NTP).
- ✓ As a network grows, it becomes difficult to ensure that all infrastructure devices are operating with synchronized time using the manual method.
- ✓ A better solution is to configure the NTP on the network. This protocol allows routers on the network to synchronize their time settings with an NTP server, which provides more consistent time settings. NTP can be set

up to synchronize to a private master clock, or it can synchronize to a publicly available NTP server on the internet.

R57. Define hypervisor. What is the difference between a Type 1 and Type 2 hypervisor? / Explain the differences between Type 1 (runs directly on hardware) and Type 2 (runs on an operating system) hypervisors, with examples.

- ✓ The hypervisor is a program, firmware, or hardware that adds an abstraction layer on top of the physical hardware. The abstraction layer is used to create virtual machines which have access to all the hardware of the physical machine such as CPUs, memory, disk controllers, and NICs.
- ✓ Type 1 hypervisor: the hypervisor is installed directly on the server or networking hardware. Then, instances of an OS are installed on the hypervisor. it has direct access to the hardware resources. They are more efficient than hosted architectures. Type 1 hypervisors improve scalability, performance, and robustness.
- ✓ Examples include: VMware vSphere, Microsoft Hyper-V, Citrix XenServer, and Red Hat Kernel-based Virtual Machine (KVM). Type 1 hypervisors are usually used on enterprise servers and datacenter networking devices.
- ✓ Type 2: This type of hypervisor (for example, VMware Fusion) requires a host OS to run. This is the type of hypervisor that is typically used by client devices. Type 2 hypervisors are also called hosted hypervisors. A big advantage of Type 2 hypervisors is that management console software is not required.

R58. Explain how Software Defined Networking (SDN) differs from traditional networking architectures.

- ✓ The data plane components of the Traditional approach and the SDN approach identical.
- ✓ In the traditional approach, the routing algorithm determines the contents of the routers' forwarding tables. A routing algorithm runs in each and every router and both forwarding and routing functions are contained within a router. The routing algorithm function in one router communicates with the routing algorithm function in other routers to compute the values for its forwarding table.
- ✓ In the SDN approach; control- plane routing functionality is separated from the physical router—the routing device performs forwarding only, while the remote controller computes and distributes forwarding tables.
- ✓ The remote controller might be implemented in a remote data center with high reliability and redundancy and might be managed by the ISP or some third party.

R59. What is the role of the SDN controller? Explain the function of the SDN controller and how it interacts with network devices (switches, routers) and applications.

- ✓ SDN Controller maintain network state information. It interacts with network control applications above via northbound API and interacts with network switches below via southbound API. It implemented as distributed system for performance, scalability, fault-tolerance, and robustness.

R60. Explain the Data Plane (responsible for forwarding packets) and Control Plane (responsible for network decision-making) in SDN.

- ✓ In generalized data plane forwarding packets, a match-plus-action table generalizes the notion of the destination-based forwarding table. The destination-based forwarding as the two steps of looking up a destination IP address (“match”), then sending the packet into the switching fabric to the specified output port (“action”). Each packet switch contains a match-plus-action table that is computed and distributed by a remote controller.
- ✓ The control plane (SDN) approach is at the heart of software-defined networking (SDN), where the network is “software-defined” because the controller that computes forwarding tables and interacts with routers is implemented in software.

R61. Why automation is important in networking?

- ✓ Automation in networking is crucial for several reasons:
 - ✓ Time saving: network issues can be detected more quickly with automated systems.
 - ✓ Accuracy: automation ensures that tasks are executed in a consistent manner, reducing human error due to manual configurations.
 - ✓ Scalability: Manual management becomes impractical when networks grow in size and complexity. Automation helps manage larger networks more efficiently.
 - ✓ Improved security: automated processes can apply security patches, update firewall configurations.
 - ✓ Better Network Analytics: Automated tools can collect data continuously and provide real-time.